

Localized Multi-Agent Debate in Small Language Models: A Judge–Proposer–Critic Framework

Hasnain Ali, Ahmed Ateeb, and Abdullah Azeem

Department of Computer Science, National University of Computer and Emerging Sciences

{i221335,i221079,i221186}@nu.edu.pk

Abstract. Small language models are attractive for privacy-sensitive legal assistants because they can be deployed locally, but their small capacity makes them vulnerable to hallucination, shallow evidence use, and unstable judgments. This paper presents a localized multi-agent debate framework for improving legal reasoning in small language models without relying on frontier-scale backbones. The system assigns three roles—Proposer, Critic, and Judge—and compares three inference configurations: a single-agent baseline, a self-critique debate, and a multi-proposer debate. The backend is implemented with FastAPI and supports structured event logging, position swapping, summary relay, LegalBench-RAG retrieval, and metric computation; the frontend provides a Streamlit dashboard for running and inspecting experiments. Evaluation is performed on legal-reasoning prompts and Knights-and-Knaves consistency tasks across four small models: LFM 350M, Qwen 0.5B, Llama 3.2 1B, and LFM 1.2B. Across 600 completed runs, the multi-proposer configuration obtains the strongest overall results, improving legal consensus from 66.0 to 76.6, Knights-and-Knaves accuracy from 54.0% to 69.1%, and reducing self-preference bias from 31.2 to 14.8 relative to the single-agent baseline. These gains come with higher latency, but the results show that localized debate can provide a practical accuracy–reliability trade-off for small legal reasoning models.

Keywords: Small Language Models · Multi-Agent Debate · Legal Reasoning · Retrieval-Augmented Generation · Self-Critique

1 Introduction

Legal question answering and contract reasoning require models to combine long-context understanding, careful evidence grounding, and stable decision making. Large language models have shown strong general reasoning abilities, but legal settings often impose privacy, cost, and deployment constraints that make cloud-hosted frontier models unsuitable. Small language models (SLMs) are therefore appealing because they can run locally and expose less sensitive data to external services. However, SLMs also have lower capacity, which often produces incomplete legal arguments, unsupported claims, brittle output formats, and inconsistent answers under small prompt changes.

This work studies whether structured interaction between localized SLM agents can compensate for these weaknesses. Rather than treating legal reasoning as a single completion, the proposed system decomposes the task into adversarial roles. A Proposer produces an initial legal argument, a Critic challenges unsupported reasoning, and a Judge synthesizes the exchange into a final verdict and consensus score. The same orchestration layer can operate as a single-agent baseline, a self-critique configuration, or a multi-proposer debate with multiple candidate arguments. The research gap addressed here is practical: existing debate and retrieval-augmented generation work often assumes large models or evaluates only final-answer accuracy, while this project evaluates debate as an engineering framework for smaller local models using consensus, faithfulness, format adherence, latency, self-preference bias, and logical consistency.

The main contributions of this paper are: (i) a localized Judge–Proposer–Critic architecture for legal reasoning with SLMs, including FastAPI orchestration, Streamlit experiment control, RAG integration, and position-swap evaluation; (ii) a comparative study of single-agent, self-critique, and multi-proposer debate across four small models; and (iii) an analysis of the quality–latency trade-off showing that multi-proposer debate improves consensus and consistency while introducing predictable inference overhead.

2 Literature Review

Retrieval-augmented generation (RAG) is widely used to reduce hallucination by conditioning language-model outputs on external documents. Lewis et al. introduced RAG as a sequence-to-sequence framework that combines parametric generation with non-parametric retrieval, demonstrating that retrieval can improve knowledge-intensive NLP tasks [5]. LegalBench-RAG extends this idea to legal contract understanding by providing deterministic retrieval evaluation against character-level legal snippets [7]. These benchmarks are important for legal AI because they evaluate whether a system retrieves and uses relevant contract evidence rather than merely producing fluent text.

Legal NLP datasets motivate the need for careful evidence grounding. ContractNLI frames contract review as document-level natural language inference, emphasizing whether legal hypotheses are entailed, contradicted, or not mentioned in agreements [4]. CUAD provides expert-labeled contract clauses for legal due diligence and highlights the difficulty of clause-level extraction [2]. MAUD extends contract understanding to merger agreements with expert annotations for complex legal provisions [9]. PrivacyQA similarly shows that real legal and policy questions require locating relevant spans in long documents rather than generating generic answers [8]. Together, these resources indicate that legal reasoning systems must be evaluated on grounding, recall, and interpretability, not only answer fluency.

Multi-agent and self-critique methods offer another route for improving reasoning. Debate-style approaches ask models to expose weaknesses in each other’s arguments so a judge can select a stronger answer [3]. Self-refinement methods

show that a model can iteratively critique and revise its own output without additional training [6]. Constitutional and critique-based alignment approaches also use feedback loops to improve helpfulness and reduce harmful outputs [1]. However, these methods are often demonstrated with large models; their reliability under local SLM constraints is less clear.

Reasoning-oriented prompting methods are also relevant. Chain-of-thought prompting improves multi-step reasoning by encouraging intermediate explanations [10], while ReAct combines reasoning traces with tool actions for information-seeking tasks [11]. The present work differs from these studies by making the interaction structure explicit: separate agents are assigned separate roles, evidence retrieval can be attached to specific turns, and the judge is evaluated under position swapping to estimate bias. The framework therefore combines ideas from RAG, critique, debate, and tool-augmented reasoning into a reproducible local system for SLM-based legal reasoning.

3 Materials and Methods

3.1 Dataset and Dataset Preparation

The study uses two evaluation sources. The first is a legal-reasoning benchmark derived from the project topics and LegalBench-RAG-style contract questions. The legal topics cover non-disclosure agreement purpose restrictions, confidentiality obligations, contract interpretation, evidence sufficiency, and related clause-level reasoning. The second source is a Knights-and-Knaves (KNK) logical-consistency benchmark used to test whether models preserve constraints across adversarial reasoning steps. Each completed run stores the topic identifier, configuration, model name, number of rounds, number of proposers, consensus score, faithfulness, format adherence, self-preference-bias score, latency, and KNK correctness.

Legal evidence is prepared through a RAG service that indexes contract text into retrievable chunks and returns structured references for the agents. In LegalBench-RAG, each benchmark item consists of a query and one or more gold snippets identified by file path and character range. The implementation uses this structure to compare retrieved evidence against expected legal references. For debate experiments, the raw run table contains 600 completed runs: 200 single-agent runs, 200 self-critique runs, and 200 multi-proposer runs. Table 1 summarizes the experimental data distribution used in the final analysis.

Table 1. Experimental data distribution.

Configuration	Runs	Models	Legal topics	KNK items/model
Single agent	200	4	5	40
Self-critique	200	4	5	40
Multi-proposer	200	4	5	40

3.2 Methodology

The proposed framework contains three agent roles. The Proposer receives a legal topic and generates a structured argument with a thesis, supporting points, and citations when retrieval is enabled. The Critic receives the Proposer’s argument and identifies weaknesses, unsupported claims, logical fallacies, or missing evidence. The Judge receives both sides and produces a verdict with a consensus score from 0 to 100. In the multi-proposer setting, two proposers generate alternative arguments before critique and judgment, increasing diversity before the final synthesis.

Let x be the legal topic, $R(x)$ be retrieved evidence, P_i be the response of proposer i , C be the critic response, and J be the judge output. A simplified two-proposer debate can be written as

$$P_i = M_i(x, R(x), h_{t-1}), \quad C = M_c(x, P_1, P_2, R(x)), \quad (1)$$

$$J = M_j(x, P_1, P_2, C), \quad s = \text{score}(J) \in [0, 100], \quad (2)$$

where h_{t-1} denotes the previous round summary and s is the extracted consensus score. In self-critique, P_1 and C are generated by one model role sequence; in the single-agent baseline, the critic stage is removed and the judge evaluates only the direct response.

The backend is implemented with FastAPI. Core modules define agent prompts, debate configuration, event emission, metric tracking, RAG access, and experiment orchestration. The system supports baseline inference, ReAct-style search, naive RAG, active RAG, and hybrid search-plus-RAG modes, although the final reported study focuses on the three debate configurations available in the result tables. The frontend is implemented in Streamlit and allows users to select models, debate modes, number of rounds, validation options, and experiment batches. Results are exported as CSV files and rendered into figures for analysis.

4 Experimental Results

4.1 Experimental Setup

Experiments compare four SLMs: LFM 350M, Qwen 0.5B, Llama 3.2 1B, and LFM 1.2B. Each model is evaluated under three configurations: single-agent, self-critique, and multi-proposer. The implementation stack consists of Python, FastAPI for the backend API, Streamlit for the frontend dashboard, SQLite for session persistence, sentence-transformer embeddings for similarity-based metrics, and a LegalBench-RAG retrieval layer. The experiments were executed as local benchmark batches and exported to `results/benchmark_full_study_results.csv`. Figures were generated from the exported result tables in the `results/figures` directory.

The main hyperparameters are the number of rounds, the number of proposers, maximum generation length, and role prompts. Single-agent runs use one proposer and no adversarial critique. Self-critique runs use one proposer

followed by a critic and judge. Multi-proposer runs use two proposers, a critic, and a judge. Experiments include one-round and two-round variants to measure whether iterative critique improves final quality. No model fine-tuning is performed; all improvements are produced by orchestration and retrieval-time prompting.

4.2 Evaluation Metrics

The primary legal metric is the judge consensus score, which measures the final judge’s confidence and agreement after considering the argument and critique. KNK accuracy measures exact correctness on logical-consistency puzzles and is reported as a percentage. Faithfulness estimates whether argument sentences are grounded in retrieved or searched evidence. The implementation first checks for explicit evidence markers and otherwise computes sentence-embedding similarity between generated claims and evidence sentences.

Format adherence measures whether outputs satisfy the required structure. Latency is measured in seconds from run start to completion. Self-preference bias (SPB) is computed through position swapping: if s_n is the consensus score in the normal order and s_w is the score after swapping positions, then

$$\text{SPB} = |s_n - s_w|. \quad (3)$$

Lower SPB indicates that the judge is less sensitive to answer ordering or authorship position. For round-to-round diversity, the system also computes information gain using cosine dissimilarity between consecutive responses:

$$\text{IG}(a, b) = 1 - \frac{\phi(a) \cdot \phi(b)}{\|\phi(a)\| \|\phi(b)\|}, \quad (4)$$

where $\phi(\cdot)$ is a sentence embedding function.

4.3 Results and Analysis

Table 2 gives the aggregate comparison across all models. Multi-proposer debate is the strongest configuration, reaching 76.6 legal consensus and 69.1% KNK accuracy. Compared with the single-agent baseline, this is a gain of 10.6 consensus points and 15.1 KNK accuracy points. Self-critique also improves over the baseline, but by a smaller margin. Format adherence remains stable across configurations, which suggests that the improvement is not caused by easier formatting but by stronger reasoning and critique.

Figure 1 visualizes the same trend. The single-agent baseline is consistently weaker, self-critique provides a middle point, and multi-proposer debate produces the highest quality. This pattern supports the central hypothesis that role separation helps SLMs expose and correct weak reasoning. The KNK improvement is especially important because these puzzles do not depend on legal phrasing; they show that debate also improves general constraint consistency.

Table 2. Overall results by configuration. Higher is better except latency and SPB.

Configuration	Runs	Consensus	KNK	Faith.	Format	Latency	SPB
Single agent	200	66.0	54.0	0.6	93.4	47.3	31.2
Self-critique	200	72.2	64.0	0.7	93.4	75.8	20.8
Multi-proposer	200	76.6	69.1	0.7	93.3	89.6	14.8

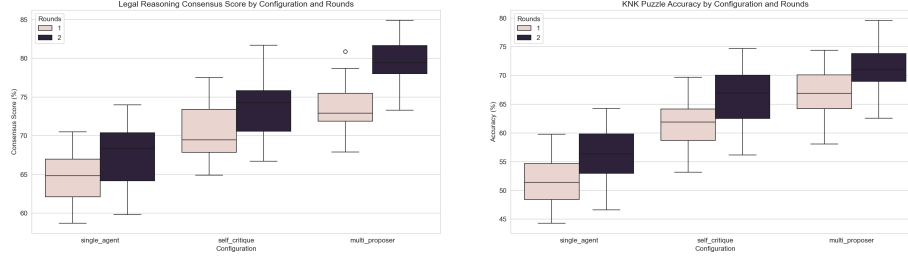
**Fig. 1.** Legal consensus and KNK accuracy by debate configuration.

Table 3 reports model-level results. LFM 1.2B is the best model overall, reaching 79.3 consensus and 74.0% KNK accuracy under multi-proposer debate. Llama 3.2 1B is competitive in consensus but has higher latency. Qwen 0.5B and LFM 350M also benefit from debate, showing that the method is useful even below one billion parameters. The relative ranking indicates that orchestration does not erase model-capacity differences, but it raises the performance floor for all tested SLMs.

Table 3. Model comparison by configuration.

Model	Configuration	Consensus	KNK	Faith.	Latency
LFM 1.2B	Multi-proposer	79.3	74.0	0.8	96.6
LFM 1.2B	Self-critique	76.9	68.6	0.8	82.7
LFM 1.2B	Single agent	69.2	58.5	0.7	51.7
LFM 350M	Multi-proposer	73.3	64.5	0.7	63.0
LFM 350M	Self-critique	68.3	58.8	0.7	52.7
LFM 350M	Single agent	62.1	49.2	0.6	32.9
Llama 3.2 1B	Multi-proposer	77.9	70.0	0.8	122.6
Llama 3.2 1B	Self-critique	73.3	66.2	0.7	102.7
Llama 3.2 1B	Single agent	68.1	55.7	0.7	63.7
Qwen 0.5B	Multi-proposer	75.8	67.9	0.7	76.0
Qwen 0.5B	Self-critique	70.2	62.7	0.7	65.3
Qwen 0.5B	Single agent	64.5	52.5	0.6	41.0

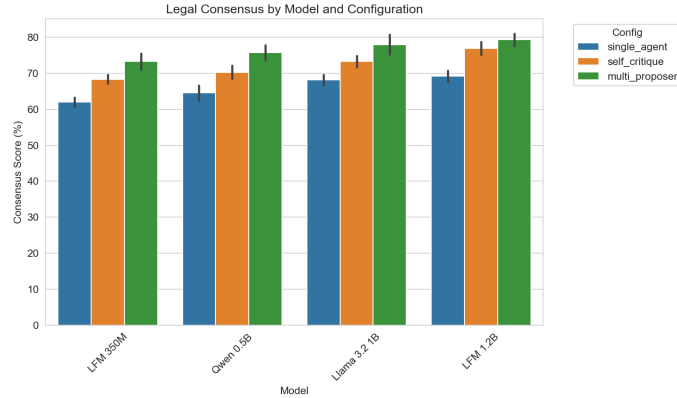


Fig. 2. Legal consensus by model and configuration.

The round analysis in Table 4 shows that two rounds improve all configurations, but the cost is substantial. Multi-proposer debate increases from 73.6 to 79.5 consensus and from 66.8% to 71.4% KNK accuracy when moving from one to two rounds, while latency rises from 62.4 to 116.7 seconds. Therefore, additional rounds are useful when answer quality is more important than response time, but one-round debate may be preferable for interactive systems.

Table 4. Effect of debate rounds.

Rounds	Configuration	Consensus	KNK	Latency
1	Single agent	64.6	51.6	32.5
1	Self-critique	70.5	61.8	52.5
1	Multi-proposer	73.6	66.8	62.4
2	Single agent	67.3	56.3	62.2
2	Self-critique	73.8	66.3	99.2
2	Multi-proposer	79.5	71.4	116.7

Bias and faithfulness results further clarify the trade-off. Multi-proposer debate reduces SPB by 68.8% compared with the single-agent baseline, while self-critique reduces it by 56.1%. Figure 4 shows that faithfulness improves with critique-based configurations and that SPB drops when the judge observes competing arguments. This supports the use of position-swapped judging as a diagnostic for legal systems: a reliable judge should not strongly prefer an argument only because of presentation order.

Finally, Figure 5 summarizes the latency-quality relationship. Debate is not free: self-critique has a $1.6\times$ latency multiplier over the single-agent baseline, and multi-proposer debate has a $1.89\times$ multiplier. However, the multi-proposer

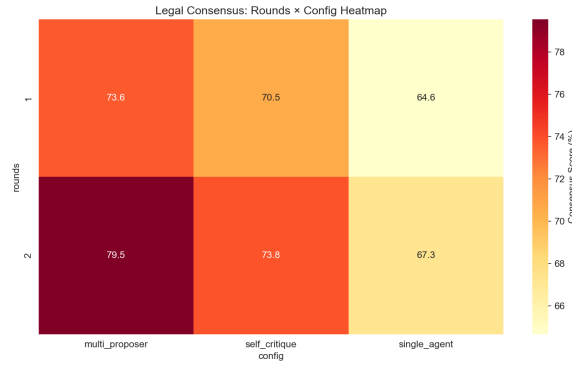


Fig. 3. Round effect heatmap for quality and runtime metrics.

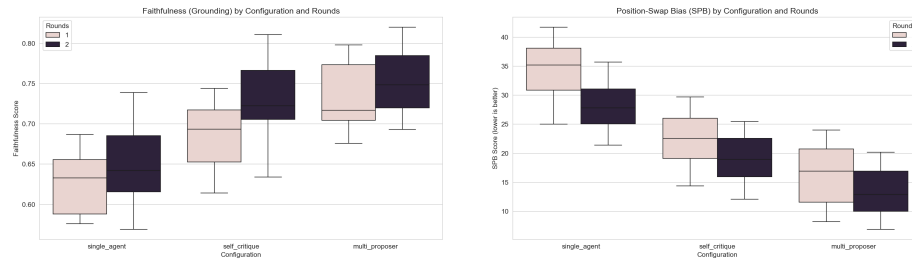


Fig. 4. Faithfulness and self-preference bias by configuration.

setting offers the best improvement per reliability objective because it increases consensus and KNK accuracy while producing the lowest SPB. For legal drafting and research support, this is a reasonable trade-off when the system is used for careful analysis rather than instant chat.

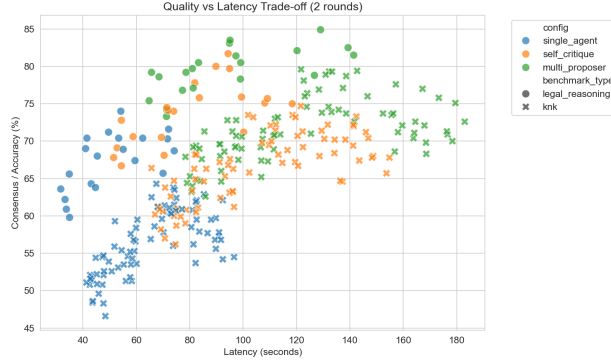


Fig. 5. Latency–quality trade-off across configurations.

5 Conclusion

This paper presented a localized Judge–Proposer–Critic framework for improving legal reasoning in small language models. The results show that structured debate is more effective than direct single-agent generation: multi-proposer debate achieved the best overall performance, raising legal consensus from 66.0 to 76.6, improving KNK accuracy from 54.0% to 69.1%, and reducing self-preference bias from 31.2 to 14.8. Self-critique also improved quality, but the strongest and most stable results came from exposing the judge to multiple proposed arguments before critique and synthesis. The main limitation is latency, since additional agents and rounds nearly double runtime compared with the baseline. Therefore, the framework is best suited to legal research and document-analysis settings where reliability matters more than instantaneous response. Future work should evaluate larger legal corpora, add stronger retrieval-grounding checks, test human legal-expert judgments, and explore adaptive stopping so the system can use extra debate rounds only when the first answer is uncertain.

References

1. Bai, Y., Kadavath, S., Kundu, S., Askell, A., Kernion, J., Jones, A., Chen, A., Goldie, A., Mirhoseini, A., McKinnon, C., et al.: Constitutional ai: Harmlessness from ai feedback. arXiv preprint arXiv:2212.08073 (2022)

2. Hendrycks, D., Burns, C., Chen, A., Ball, S.: Cuad: An expert-annotated nlp dataset for legal contract review. arXiv preprint arXiv:2103.06268 (2021)
3. Irving, G., Christiano, P., Amodei, D.: Ai safety via debate. arXiv preprint arXiv:1805.00899 (2018)
4. Koreeda, Y., Manning, C.D.: Contractnli: A dataset for document-level natural language inference for contracts. arXiv preprint arXiv:2110.01799 (2021)
5. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems* **33**, 9459–9474 (2020)
6. Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegrefe, S., Alon, U., Dziri, N., Prabhume, S., Yang, Y., et al.: Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems* **36** (2023)
7. Pipitone, N., Houir Alami, G.: Legalbench-rag: A benchmark for retrieval-augmented generation in the legal domain. arXiv preprint arXiv:2408.10343 (2024)
8. Ravichander, A., Black, A.W., Wilson, S., Norton, T., Sadeh, N.: Question answering for privacy policies: Combining computational and legal perspectives. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*. pp. 4949–4959. Association for Computational Linguistics (2019)
9. Wang, S.H., Scardigli, A., Tang, L., Chen, W., Levkin, D., Chen, A., Ball, S., Woodside, T., Zhang, O., Hendrycks, D.: Maud: An expert-annotated legal nlp dataset for merger agreement understanding. arXiv preprint arXiv:2301.00876 (2023)
10. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. In: *Advances in Neural Information Processing Systems*. vol. 35, pp. 24824–24837 (2022)
11. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: React: Synergizing reasoning and acting in language models. In: *International Conference on Learning Representations* (2023)